Hummingbirds and swifts

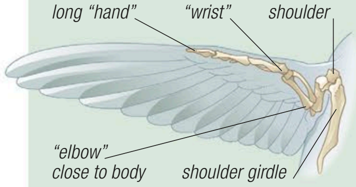
PHYLUM	Chordata
CLASS	Aves
ORDER	Apodiformes
FAMILIES	4
SPECIES	470

Swifts and hummingbirds share a unique wing structure that makes them acrobatic fliers, capable of highly intricate aerial maneuvers. However, their appearance and lifestyles differ greatly. The soberly colored swifts rarely land, spending their lives in midair in search of flying invertebrates; they can sleep, and even mate on the wing. The multicolored hummingbirds hover around flowers to feed, and perch readily. Swifts occur worldwide, but hummingbirds are restricted to the Americas.

FEEDING ON NECTAR
When feeding, a hummingbird (here, a green violetear from Costa Rica) hovers in front of a flower, maneuvering its bill up the tube to draw up the nectar with its long tongue.

Anatomy

Both hummingbirds and swifts have a compact, muscular body, and relatively small feet. Although swifts are dull in color, hummingbirds are remarkable for their dazzling colors and patterns. Hummingbirds have a specialized bill designed for removing nectar from flowers. The bill length and shape are variable, and often match the shape of the flowers at which the birds feed. Swifts have a small bill with a wide gape for trapping tiny insects in flight. The hummingbird family contains many of the world's smallest birds.



WINGS
In hummingbirds and swifts, the joint between the upper and lower arm, the "elbow" is very close to the body, giving the wings great flexibility and leverage. This feature enables hummingbirds to hover.

Flight

Hummingbirds beat their wings in a figure-eight pattern, which allows them great maneuverability; they are the only birds that can fly backward, and even upside-down. Smaller species may beat their wings 80 times a second. Swifts do not hover, but can vary the speed of their wingbeats to turn sharply.

SWIFT IN FLIGHT
wifts resemble swallows, but are not closely related to them. They spend more time in the air, and are quicker, more erratic fliers. Swallows perch more readily than swifts.

Eutoxeres aquila

White-tipped sicklebill

Location S. Central America to N.W. South America

Length 4¾–5½ in (12–14 cm)
Weight ⅜–7/16 oz (10–13 g)
Plumage Sexes alike
Migration Nonmigrant
Status Least concern

This large hummingbird is green above with streaked underparts, and has white-tipped tail feathers and an unmistakable downcurved bill. The shape of the bill allows it to drink nectar from flowers that have curved throats, especially heliconias. Unlike most hummingbirds, it perches on the flower while feeding from it. It also gleans spiders from their webs and catches insects on the wing. The female lays 2 elliptical eggs in a cup-shaped nest, which is loosely woven from palm fibers and attached by cobwebs to the tip of a hanging leaf.

Eriocnemis isabellae

Gorgeted puffleg

Location N.W. South America (Andes)

Length 4 in (10 cm)
Weight 1/10–¾ oz (3.9–4.5 g)
Plumage Sexes differ
Migration Nonmigrant
Status Critically endangered

Discovered in 2005, this hummingbird is restricted to cold, humid elfin forest (just 19–26 ft/6–8 m high) in a small region—scarcely (9 square miles/ 10 square km) in area—dominated by rocky outcrops high in the Colombian Andes. It is threatened with extinction by loss of habitat. It has a large, brilliantly colored throat patch with a violet-blue center and iridescent green edges. Known as a gorget, this throat patch along with the white thigh tufts gives rise to its common name.

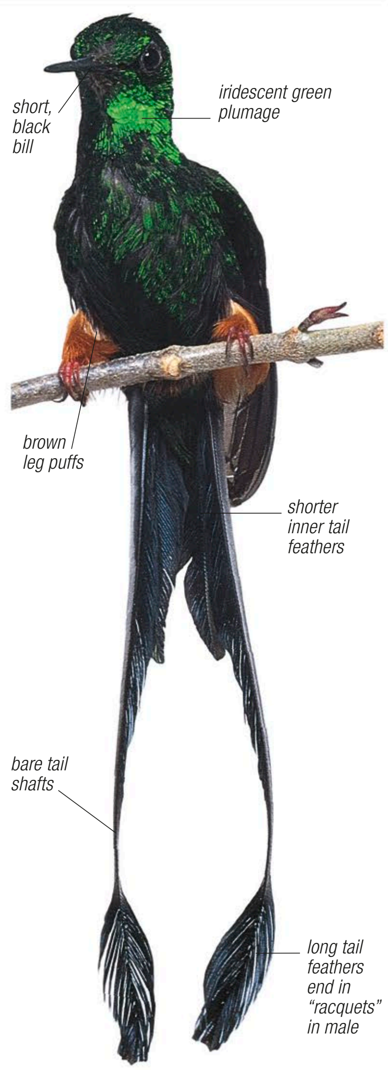
Ocreatus underwoodii

Booted racket-tail

Location N.W. and W. South America (Andes)

Length 4¼–6 in (11–15 cm)
Weight ⅛ oz (3 g)
Plumage Sexes differ
Migration Nonmigrant
Status Least concern

Elongated tail feathers with bare shafts ending in blue-black "racquets" distinguish the male from the female of this tiny species. When courting, the male puts on a spectacular flight display, holding up the fluffy white or brown "puffs" or leg feathers, and flicking his tail feathers up and down to produce a whiplike sound. The male and female fly a regular route when foraging for food, calling to one another to stay in contact. The bird hovers in front of a flower, its rapidly beating wings making a distinct humming noise, and pierces the flower tube with its bill to reach the nectar. Like most hummingbirds, it often becomes torpid at night, its body temperature dropping almost to that of the air in order to save energy. Hummingbirds are so tiny that they would risk starving to death if they maintained their normal body heat throughout the night.



BIRDS